Louvain Community Detection

Student Social Network — Neo4j GDS · GDS-43

1. Objective

This document presents the results of applying the Louvain algorithm to the student friendship network. The Louvain algorithm detects communities by iteratively maximizing the modularity score Q, which measures the density of connections inside communities compared to connections between communities. A modularity above 0.30 is generally considered a significant community structure.

2. Implementation

Graph Projection

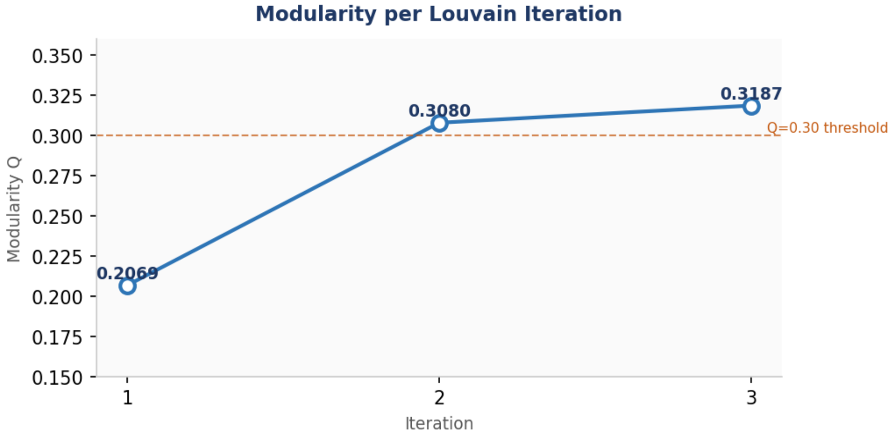
```
CALL gds.graph.project(
    'student-friendship-graph',
    'Student',
    { EST_AMI_DE: { orientation: 'UNDIRECTED' } }
)
YIELD graphName, nodeCount, relationshipCount;
```

Louvain Write

```
CALL gds.louvain.write(
    'student-friendship-graph',
    { writeProperty: 'louvainCommunity' }
)
YIELD communityCount, modularity, modularities, nodePropertiesWritten;
```

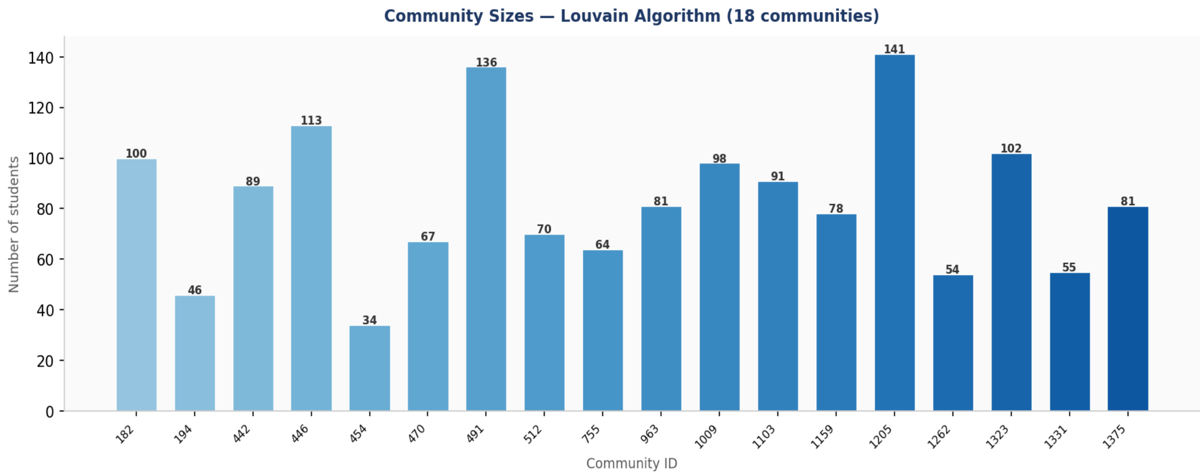
3. Global Results

Metric	Value
Number of communities	18
Final modularity Q	0.3187
Nodes written	1,500
Iterations	3
Modularity — iteration 1	0.2069
Modularity — iteration 2	0.3080
Modularity — iteration 3	0.3187



4. Community Size Analysis

The 18 communities vary significantly in size. The largest community (1205) contains 141 students while the smallest (194) has only 46. This size variation is typical of real-world social networks where a few large clusters coexist with several smaller, tighter groups.

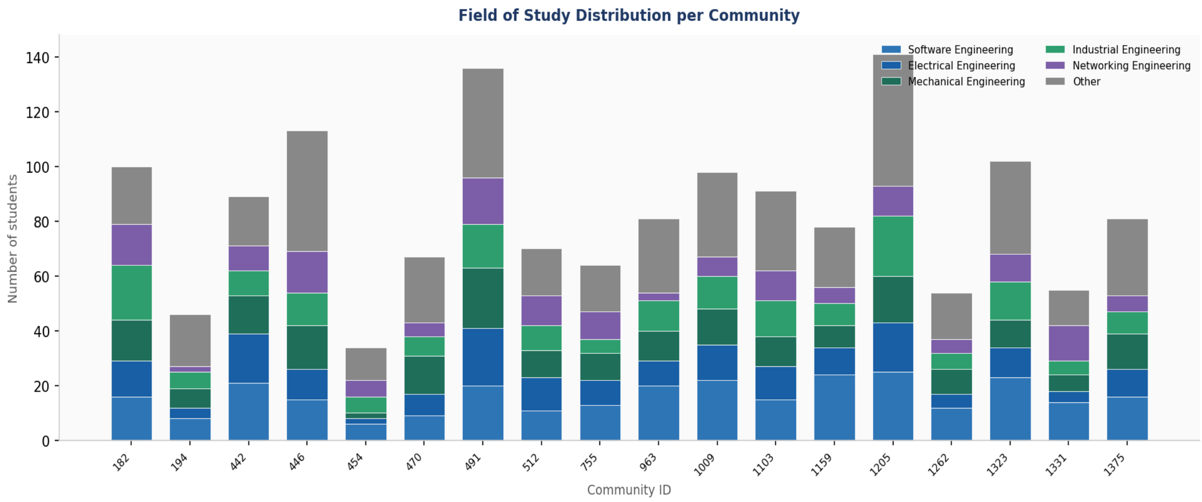


Community ID	Size	% of total
182	100	6.7%
194	46	3.1%
442	89	5.9%
446	113	7.5%
454	34	2.3%
470	67	4.5%
491	136	9.1%
512	70	4.7%
755	64	4.3%
963	81	5.4%
1009	98	6.5%

1103	91	6.1%
1159	78	5.2%
1205	141	9.4%
1262	54	3.6%
1323	102	6.8%
1331	55	3.7%
1375	81	5.4%

5. Field of Study Distribution per Community

The stacked chart below shows the composition of each community by field of study. If the Louvain algorithm had detected field-based communities, each bar would be dominated by a single color. The mixed distribution observed confirms that communities are formed by social connections, not by academic department.



Field distribution — top 6 largest communities

Community	Software Eng.	Electrical Eng.	Mechanical Eng.	Industrial Eng.	Networking Eng.	Other	Total
1205	25	18	17	22	11	48	141
491	20	21	22	16	17	40	136
446	15	11	16	12	15	44	113
1323	23	11	10	14	10	34	102
182	16	13	15	20	15	21	100
1009	22	13	13	12	7	31	98

6. Observations & Analysis

• Modularity above threshold

The final modularity Q = 0.3187 exceeds the 0.30 threshold considered significant in the literature. This confirms that the Louvain algorithm has detected a genuine community structure in the student friendship network, not a random partition.

• Convergence in 3 iterations

The algorithm converged in only 3 iterations, going from Q=0.2069 to Q=0.3080 to Q=0.3187. The largest gain happened between iteration 1 and 2 (+0.10), while the final refinement was smaller (+0.009), indicating stable convergence.

• Communities are not field-based

Every community contains students from all 6 fields of study in roughly similar proportions. This is the most important observation: the Louvain algorithm grouped students by social connections, not by academic department. This means friendships in this network cross departmental boundaries freely.

• 'Other' dominates every community

The 'Other' category (30% of students by design) appears as the largest or second-largest group in almost every community. This is expected given its size in the original dataset — it is not a signal of algorithmic bias.

• High size variance

Community sizes range from 46 (community 194) to 141 (community 1205). Three communities — 1205, 491, 446 — account for 34% of all students. This skewed distribution suggests a core-periphery structure where a few large social hubs anchor the biggest communities.

• Community IDs are internal node IDs

The community IDs (182, 194, 442...) are not sequential labels — they are the internal Neo4j node IDs of the student chosen as the representative of each community by the Louvain algorithm. This is normal GDS behavior.

• Recommendation system implication

Since communities are socially formed and not field-based, the recommendation system can safely use community membership as a feature to recommend new friends without inadvertently creating field-based filter bubbles.

7. Conclusion

The Louvain algorithm successfully partitioned the 1,500-student friendship network into 18 meaningful communities with a modularity of 0.3187. The communities reflect genuine social structures rather than demographic or academic groupings. These results validate the graph modeling approach and provide a solid foundation for the next steps: Leiden and Label Propagation comparison (GDS-44, GDS-45), and the integration of community membership as a feature in the recommendation system.